2 **(a)** Explain how a body moving at constant speed can experience an acceleration.

....

....

.... [2]

(b) A ball at rest at O is slightly displaced so that it slides down a frictionless track towards the loop-the-loop of radius R as shown in Fig. 2.1.

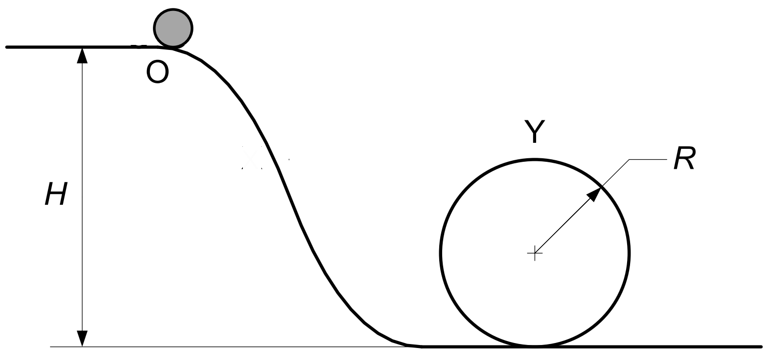

Fig. 2.1

(i) On Fig. 2.2, draw and label the forces acting on the ball when it is at the highest point Y of the loop-the-loop.

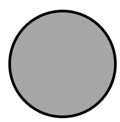

Fig. 2.2

[2]

(ii) Show that the minimum speed of the ball at point Y so that it remains in contact with the track is $\sqrt{gR}$.

[2]

(iii) Hence, determine an expression in terms of R for the minimum height H of point O with respect to the lowest point of the loop-the-loop.

[2]